ELECTRIC VEHICLE MARKET PERFORMANCE AND INSIGHTS

B.Tech Project Report and Tableau Public Dashboard Guide

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1. Abstract

This project studies electric vehicle (EV) market performance using a monthly state-wise dataset covering January 2024 to December 2025. It compares sales across two-wheelers, three-wheelers and four-wheelers, examines changes over time, and identifies states with higher activity. Tableau Public is used to create an interactive dashboard for exploring the data.

Important data note: The attached dataset is synthetic and created for academic practice and Tableau learning. It is not official VAHAN, government, or industry sales data. Replace it with verified data before making real-world market claims.

2. Introduction

Electric vehicles use electric motors and rechargeable batteries instead of relying only on internal-combustion engines. EV adoption is influenced by purchase price, charging availability, battery technology, operating cost, government policy and consumer awareness. Studying sales patterns helps explain which vehicle categories and regions show stronger demand.

3. Objectives

- Analyze EV sales trends by month, year and quarter.
- Compare two-wheeler, three-wheeler and four-wheeler sales.
- Identify states with the highest and lowest sales in the sample.
- Build interactive Tableau visualizations and filters.
- Present key observations and recommendations.

4. Dataset Description

The CSV and Excel files contain 576 rows: 8 states × 24 months × 3 vehicle categories. Each row represents the illustrative sales count for one state, month and vehicle type.

Field	Description
Month	Month start date (YYYY-MM-DD)
Year	Calendar year
Quarter	Calendar quarter
State	State/UT represented in the sample
Vehicle_Type	Two-Wheeler, Three-Wheeler or Four-Wheeler
EV_Sales	Illustrative monthly sales count

5. Analysis and Key Insights

The sample contains **599,889** illustrative EV sales records in aggregate across the two-year period. Category contribution is shown below; these percentages describe only this synthetic dataset.

Vehicle category	Sample sales	Share
Two-Wheeler	401,713	67.0%
Three-Wheeler	119,757	20.0%

Four-Wheeler	78,419	13.1%

In this generated sample, **Maharashtra** has the highest aggregate sales and **Madhya Pradesh** the lowest. The dataset is intentionally designed with a gradual upward trend and a seasonal pattern so that learners can practice trend analysis. Do not interpret these results as actual state rankings.

6. Tableau Public Dashboard – Steps

1. Open Tableau Public and choose Connect → To a File → Text File (CSV).
2. Select ev_market_tableau_dataset.csv. Check that Month is recognized as a Date and EV_Sales as a Number.
3. Create a line chart: Month on Columns, SUM(EV_Sales) on Rows, and Vehicle_Type on Color.
4. Create a horizontal bar chart: State on Rows and SUM(EV_Sales) on Columns; sort descending.
5. Create a category comparison chart: Vehicle_Type and SUM(EV_Sales). A bar chart is suitable.
6. Create KPI cards for total EV sales, number of states and number of vehicle categories.
7. Add filters for Year, State and Vehicle_Type. Apply filters to all relevant worksheets.
8. Place the charts and KPI cards on a dashboard. Add a clear title and a note that the dataset is synthetic.
9. Choose Publish Workbook to Tableau Public and copy the dashboard link for your submission.

7. Suggested Dashboard Layout

Top row: project title and KPI cards. Middle row: monthly sales trend line chart and vehicle-category comparison. Bottom row: state-wise sales bar chart and filters. Use consistent number formatting, readable labels and a simple colour palette.

8. Findings (for this sample)

The dashboard should make it easy to compare category contribution, identify month-to-month movement and rank states. Because the dataset is simulated, the correct academic conclusion is that Tableau can reveal patterns in structured EV data; no factual conclusion about India's real EV market should be drawn from this practice dataset.

9. Conclusion

This project demonstrates a complete basic data-analysis workflow: preparing a tabular dataset, connecting it to Tableau Public, creating visualizations, adding interactive filters and communicating findings. The same dashboard design can be reused with official EV registration or sales data for a real market study.

10. Future Scope

Future work can add verified state-wise VAHAN registrations, charging-station counts, EV policy dates, fuel prices, population and vehicle-price data. These additional fields can support comparisons such as EV adoption per capita and the relationship between charging infrastructure and registrations.

11. References / Suggested Data Sources

For a real-data version, consult the VAHAN Dashboard (Ministry of Road Transport and Highways), the Ministry of Heavy Industries, the Bureau of Energy Efficiency and official state transport department publications. Record the download date, reporting period and definitions used. Confirm licensing and attribution requirements before reusing data.

Submission checklist: Attach this report, the CSV/Excel dataset, screenshots of at least three Tableau worksheets, one dashboard screenshot and the published Tableau Public URL.